

Chapter 3.

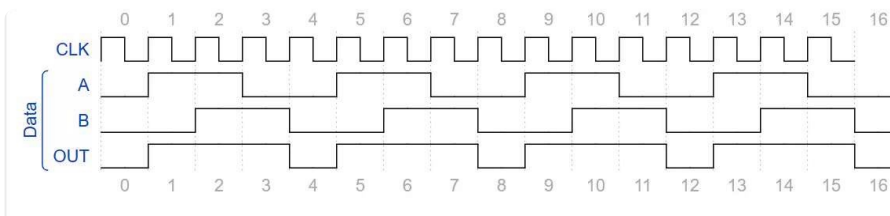
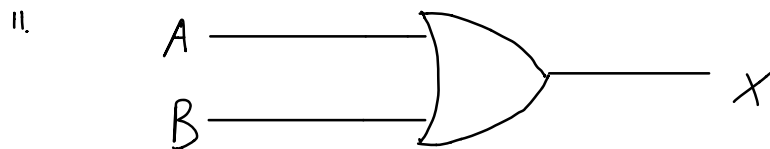
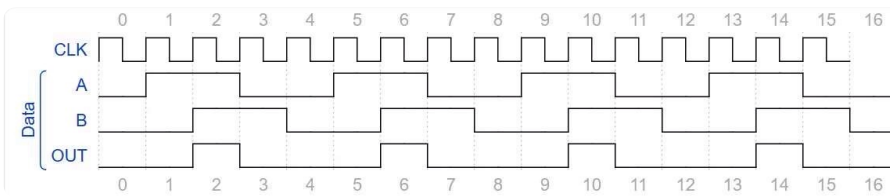
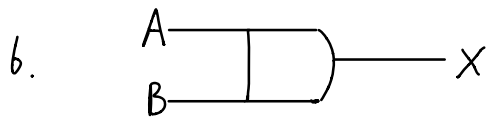
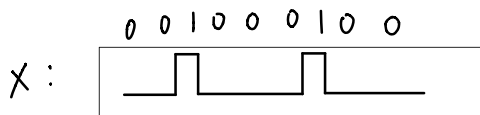
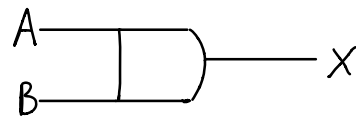
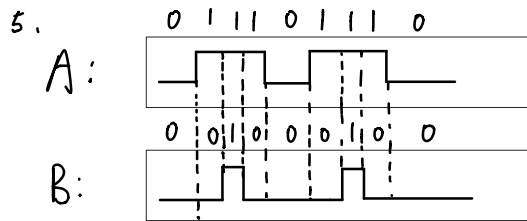
2. $F = \overline{\overline{A}} = \overline{1} = 0$

A =

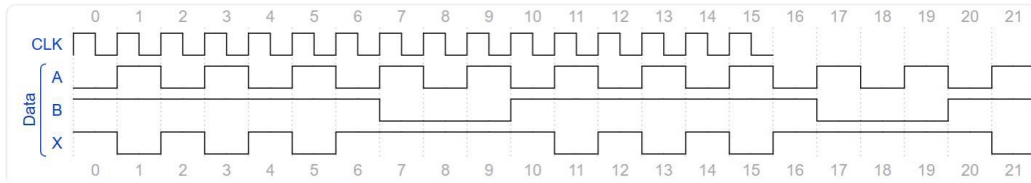
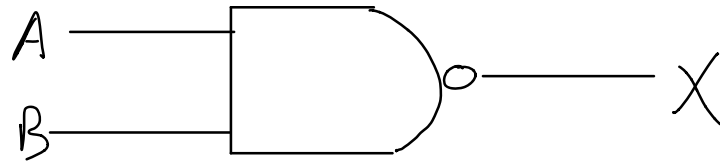
1
0

F =

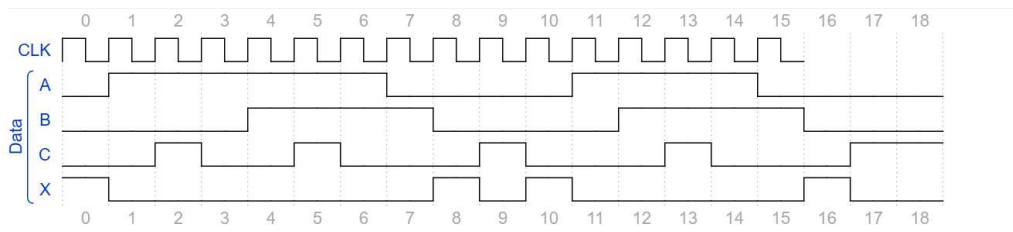
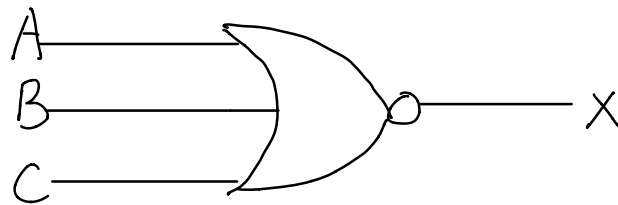
1
0



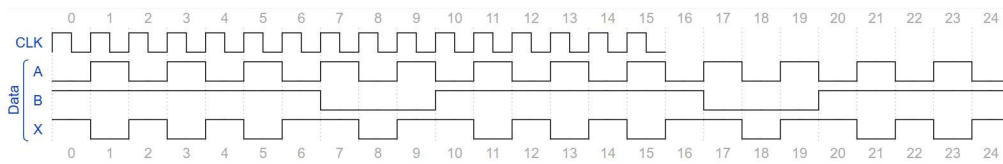
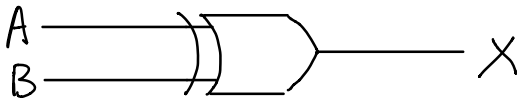
17



22.



26.



Chapter 4:

$$10. \quad a). \quad \overline{A\bar{B}(C+\bar{D})} = \overline{A\bar{B}} + \overline{(C+\bar{D})} = \bar{A}+B + \bar{C}\bar{D}$$

$$b). \quad \overline{AB(CD+EF)} = \overline{AB} + \overline{(CD+EF)} = \bar{A}+\bar{B} + \bar{C}\bar{D}\bar{E}\bar{F}$$

$$= \bar{A}+\bar{B} + (\bar{C}+\bar{D})(\bar{E}+\bar{F})$$

$$c). \quad \overline{(A+\bar{B}+C+D)} + \overline{ABCD} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}+\bar{B}+\bar{C}+\bar{D}$$

$$d). \quad \overline{(\bar{A}+B+C+D)} (\overline{A\bar{B}C\bar{D}}) = \overline{(\bar{A}+B+C+D)} + \overline{A\bar{B}C\bar{D}}$$

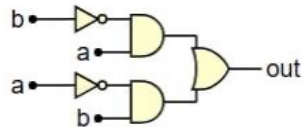
$$= (\bar{A}+B+C+D) + (A(\bar{B}+\bar{C})\bar{D})$$

$$= \bar{A}+B+C+D + (A\bar{B}\bar{D}+A\bar{C}\bar{D})$$

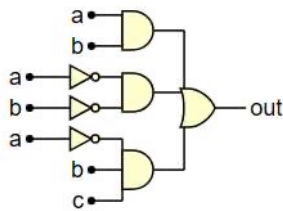
$$\begin{aligned}
 e) \quad & \overline{\overline{AB} (CD + \overline{E}F) (\overline{AB} + \overline{CD})} \\
 &= \overline{\overline{AB} (CD + \overline{E}F)} + \overline{(\overline{AB} + \overline{CD})} \\
 &= AB (\overline{CD + \overline{E}F}) + \overline{\overline{AB} + \overline{CD}} \\
 &= AB (\overline{CD} + \overline{E\overline{F}}) + AB\overline{C}\overline{D} \\
 &= AB (\overline{C} + \overline{D} + \overline{E}\overline{F}) + AB\overline{C}\overline{D} \\
 &= AB (CD + \overline{C} + \overline{D} + \overline{E}\overline{F})
 \end{aligned}$$

5.

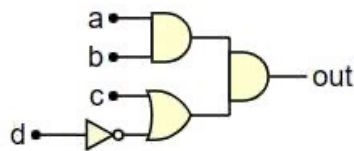
$$a). \overline{A}B + A\overline{B} \rightarrow (A \wedge \neg B) \vee (\neg A \wedge B)$$



$$b). \overline{A}B + \overline{A}\overline{B} + \overline{A}BC \rightarrow A \wedge B \vee (\neg A \wedge \neg B) \vee (\neg A \wedge B \wedge C)$$



$$c). \overline{A}B(C + \overline{D}) \rightarrow \neg A \wedge B \wedge (C \vee \neg D)$$



$$d). A + B[C + D(B + \overline{C})]$$

$$\rightarrow A \vee [B \wedge (C \vee (D \wedge (B \vee \neg C)))]$$

